Experiment 1: Additional Problem Statements

Problem Statement 1: Student Marks Analytics System

A class has N students (N ≤ 100). Store the marks of all students (0–100) in an array. Write a function that generates marks randomly in range from 0 to 100. Develop a menu-driven program to perform the following operations:

Display the highest and lowest marks.

1. Find the average marks of the class.
2. Count the number of students who scored:
3. Above average
4. Below average
5. Search whether a particular mark exists in the class (Linear Search).
6. Display the frequency of each distinct mark.

Problem Statement 2: Daily Temperature Monitoring System

A weather station records the temperature of a city for 30 consecutive days.

Store the temperatures in an array and perform the following operations:

1. Find the hottest and coldest day.
2. Find the longest continuous streak of increasing temperatures.
3. Sort the temperatures in ascending order using any sorting algorithm studied.
4. After sorting, allow the user to search for a particular temperature using Binary Search.
5. Identify how many days had temperatures higher than the monthly average.

Problem Statement 3: Inventory Management for a Retail Store

A retail shop maintains the following information for N products:

- Product ID
- Product Name
- Quantity Available

Store the data using arrays and implement the following operations:

1. Search a product using its Product ID.
2. Identify the product with the highest and lowest stock.
3. Update the stock quantity after a sale or new shipment.
4. Display all products sorted in ascending order of available quantity.
5. Display all products whose stock is below a user-defined threshold (e.g., less than 10 units).

Constraints

- Do not use structures, linked lists, or dynamic memory allocation.
- Use only arrays and functions.